

SPECIFICATION**Introduction**

The R502 Heater Delay Timer Board is an electronic heater control board which provides, in one device, control of a line voltage heater from a thermistor. The primary purpose is to accumulate the time the thermistor is between two temperatures: LoTemp and HiTemp (e.g. 25F and 40F) and when the accumulating timer reaches an initial preset number of hours (selectable 0, 3, 6, 12) the relay will energize until HiTemp is reached. Once HiTemp is reached, the relay will de-energize and select the next lower number of hours for the accumulated time. A RstTemp (e.g. 55F) sensed will reset the board to 6 hours of accumulated time. An LED indicates the accumulated time period in force at any time or a thermistor short or open.

Features

Led to indicate accumulated time necessary for relay to energize or board malfunction:

- 1 blink – 12 hours
- 2 blinks- 6 hours
- 3 blinks – 3 hours
- 4 blinks – 0 hours (instantaneous switching on/off at LoTemp and HiTemp)
- Continuous flash- Thermistor shorted or Open
- Board comes with a 10K ohm thermistor (3 ft-1 meter lead) which plugs into the screw terminals on the board.
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Electrical specification

- Nominal Input: 24 VAC
- Operating Input Range: 18-30VAC
- Fan Outputs: 12 FLA (30 LRA) at 240 VAC

PCB Dimension: 3.5" x 2.125"

Terminals

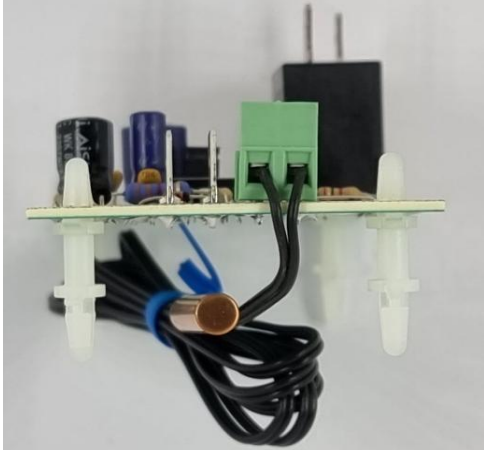
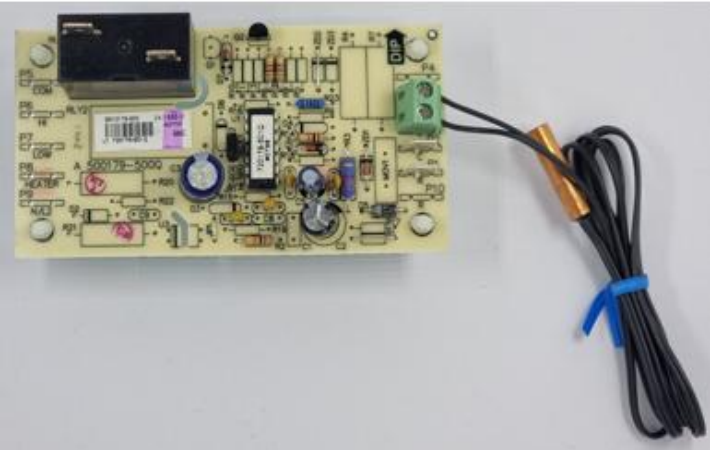
Line voltage connections are .250" x 0.032" quick connect terminals on top of the relay.

Quick connect terminals are provided for T and C inputs from the transformer.

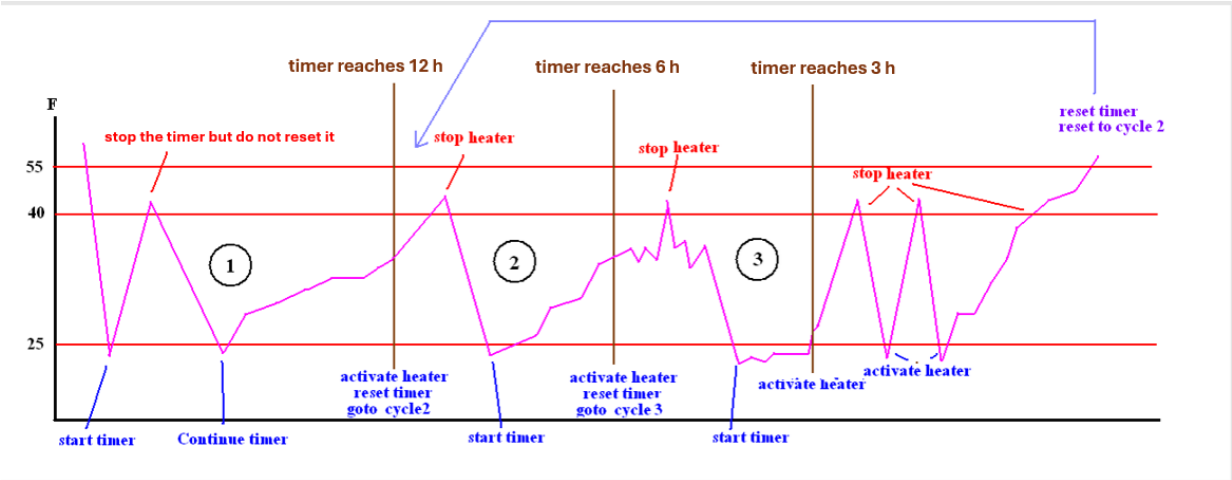
Quick connect terminals are provided for a thermostat.

- 24VAC Hot (R) (fused)
- 24VAC Neutral (C) (same terminal for transformer common input)
- Fan Output (G)

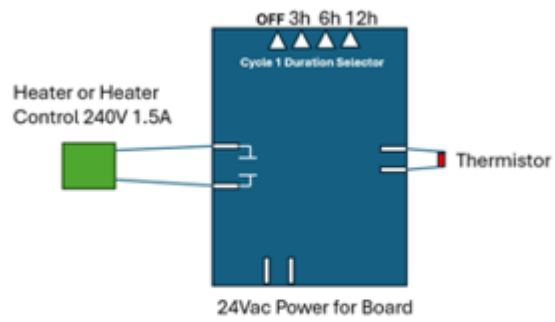
Component Layout



Example Timing Diagram: 12h selected; 25F/40F selected



Typical Wiring Diagram



R502 Temperature Regulated
Heater Timer with Diminishing
Time Cycles